Memory & Data Layer (Shared Vector Store)

DATABASE
Supabase Platform
Stores core data
Supabase Platform

DATABASE
Student Profiles (Qdrant)
Stores student identity, preferences, and long-term learning goals.
Qdrant
Vector Embeddings

LeakSentinel Workflow (Dashed Lines)

Trigger Alert (Dashed)

SERVICE
Monitor Agent (LeakSen...
Scans Telegram, Reddit, and Twitter for potential exam-related leaks.
Mastra
Playwright
Python

Analyze (Dashed)

SERVICE
Detect Agent (LeakSent...
Embeds scraped data and compares it against known leak patterns in
Mastra
Sentence-Transformers
Qdrant

SERVICE
Alert Agent (LeakSentin...
Generates confidence scores via Enkrypt and pushes alert
Mastra
Enkrypt AI Safety
Node.js

Pattern Match Res
d)

DATABASE
Weak Concepts (Qdrant)
Tracks specific topics where the student has shown low proficiency.
Qdrant
Knowledge Graph Embeddings

4. Retrieval (Green)
4. Retrieval
Usage Tracking
4. Retrieval
5: Shared Context Res
4. Retrieval (Green)

AI Orchestration Layer

SERVICE
Mastra Orchestra...
AI Tutor Orchestrator with a Proactive Resurface Loop (every 5
Mastra
TypeScript
Node.js
LangGraph-style Loops

6. Raw Response

SERVICE
Groq LLaMA 3.1
High-speed LLM inference for generating tutor responses.
Groq
LLaMA 3.1
LPUs

5. Validate Context

DATABASE
Session History (Qdrant)
Logs all past interactions for context-aware tutoring.
Qdrant
Temporal Embeddings

Auth & Metadata
5. Validate Context

7. Check Response

3. Assess Level (Green)

Enkrypt AI Safety Layer (Orange Borders)

SERVICE
Input Guard (Enkrypt)
Step 2: Screens toxicity before processing. [Orange Border Safety
Enkrypt AI
Python
Guardrails AI

Push Alert (Dashed)

7. Check Response (Green)

Legend & Key

SERVICE
Diagram Legend & Key
Green Solid: Tutoring Flow | Dashed: LeakSentinel Flow | Orange Borders:
Documentation

Scrape (Dashed)

DATABASE
Leak Patterns (Qdrant)
Stores known patterns and signatures of leaked exam papers.
Qdrant
Pattern Matching Embeddings

5. Validate Context (Green)

SERVICE
Retrieval Guard (Enkrypt)
Step 5: Validates retrieved context before use. [Orange Border Safety
Enkrypt AI
Python
Context Validation

5. Validate Context

External Sources

CLIENT
Public Sources
Public forums, social media, and educational sites monitored for
Web
Social Media
Forums

SERVICE
Output Guard (Enkrypt)
Step 7: Hallucination and bias check before delivery. [Orange Border
Enkrypt AI
Python
Hallucination Detection

2. Screen Input (Green)

8. Receive Response (Green)

API Gateway Layer

API GATEWAY
VidyaGuard Backend
Routes incoming requests
VidyaGuard Backend

1. Submit Doubt (Green)
8. Safe Response (Green)

Frontend Layer

FRONTEND
VidyaGuard Frontend
React-based student dashboard for interacting with the AI tutor and
React
Vite
Tailwind CSS
Lucide React